

Angelina Jiang

416-951-2328 | angelinajiang60@gmail.com | [linkedin.com/in/angelina-qihong-jiang](https://www.linkedin.com/in/angelina-qihong-jiang) | github.com/angelsanddevslo1

EDUCATION

University of Toronto

Sept. 2025 – May. 2028

Candidate for Bachelor's Degree, Computer Science

cGPA: 3.97/4.00

EXPERIENCE

Artificial Intelligence Research Assistant

May 2026 – Present

The Matter Lab, University of Toronto

Toronto, ON

- Built an evaluation harness to benchmark LLM code-generation agents across models and workflows, looking at token usage, latency, and code complexity to compare agent efficiency across MCP-server tool calls
- Scaled agentic pipelines beyond interactive-session limits by orchestrating multi-hour batch compute across HPC nodes with SLURM, enabling long-running autonomous workloads
- Optimized structural execution graphs in Python for the El Agente: Grafico with Prof. Alán Aspuru-Guzik

Software Engineer

May 2026 – Present

Department of Computer Science, University of Toronto

Toronto, ON

- Shipped feature enhancements to Courseography, an open-source website in Haskell and React, used by 250+ computer science students per month to visualize course prerequisites, program requirements, and term schedules
- Performed a schema migration in SQLite that refactored internal data access to improve readability
- Built and maintained a quality pipeline with Jest and Coveralls, achieving 59% statement coverage across 500+ automated tests on CircleCI by applying test-driven development (TDD) and expanding unit-test coverage

Software Engineer

May 2026 – Present

Clean Architecture Visualizer for Education

Toronto, ON

- Built features for CAVE, a program built in TypeScript, JavaScript and React, that helped 100+ users learn and practice software design principles
- Delivered a matching exercise end-to-end, owning the full cycle from Figma design to implementation and testing
- Reduced wall-clock runtime by 5.1% by adopting DevOps practices and reconfiguring GitHub Actions (CI/CD)

Software Engineer

May 2026 – Present

IQBank

Toronto, ON

- Built a full-stack user import feature in React.js and Node.js for an exam-preparation platform used by 300+ users
- Streamlined deployment and local development by containerizing the application with Docker
- Created REST API endpoints to fetch and display users from the PostgreSQL database using Prisma

PROJECTS

Agentic AI Travel App | *React Native, TypeScript, Google APIs, Gemini*

March 2026

- Built a cross-platform mobile app that adapts travel itineraries from user energy check-ins and location context
- Engineered an agentic pipeline that evaluates user state, filters nearby options via the Google Maps SDK, and recommends alternative activities with reasoning and directions
- Integrated the Gemini API to generate personalized itineraries from mood and wellness data persisted in Supabase

Instrument Classification ML Model | *pandas, NumPy, PyTorch, Google Colab, Kaggle*

Feb 2026 – April 2026

- Built and trained a convolutional neural network (CNN) that classifies musical instruments across 18 categories
- Engineered an end-to-end data pipeline that cleaned raw Kaggle .wav files and generated structured .csv metadata with pandas, then split the dataset 80/20 for training and testing
- Improved classification accuracy from 56% to 96% through hyperparameter tuning and audio data augmentation, using GPU-accelerated training in Google Colab
- Deployed the model as a full-stack web app with a Flask REST API serving a React frontend, containerized with Docker on Hugging Face Spaces, letting users upload .wav or .mp3 clips and receive instrument predictions

Flight Inflation Detector Agent | *Python, pandas, Matplotlib, NumPy*

Feb 2026 – March 2026

- Developed an XGBoost model during the SDSS Datathon that flags flight-fare inflation against expected pricing, reaching 91.8% accuracy on real-time fare data

- Ensured statistical integrity by cleaning source data with pandas and NumPy and validating with an 80/20 split and cross-validation to limit overfitting
- Communicated model insights and inflation trends through Power BI dashboards and Matplotlib visualizations

1v1 Multi-Agent RL Environment | *PyGame, PyMunk, Stable Baselines3, PyTorch* Oct 2025 – Nov 2025

- Built an autonomous agent for a 1v1 competitive environment, implementing Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN) to optimize decision strategies
- Designed reward functions and tuned hyperparameters, evaluating agent performance through repeated testing
- Deployed and benchmarked agents in a live tournament with 400+ participants

Outfit Creation App | *React Native, JavaScript, Flask, MongoDB, Expo* Aug 2025

- Built a full-stack cross-platform mobile app that lets users assemble outfits through a browse-and-swipe interface
- Designed a Flask REST API backend with user data and saved outfits persisted in MongoDB
- Implemented a Playwright web scraper integrated with SerpAPI's Google Image Search to surface e-commerce listings for similar clothing

Smile Detection Program | *Python, OpenCV* July 2025

- Developed a real-time photo capture program to detect faces and smiles using OpenCV and instantly save the picture to the local file system
- Implemented an automatic photo capture and save system that instantly stores detected images to the user's local file directory

AI Marketing Website | *HTML, CSS, Flask, JavaScript* Sept 2024

- Built a web platform that generates marketing content from user prompts using generative AI such as Cohere for captions and ModelsLab for image generation
- Automated a posting system that publishes generated content across social media platforms, such as Twitter, using Twitter's API
- Designed an intuitive user interface under hackathon time constraints

Exam Sharing Platform | *React.js, Flask, Python, SQLAlchemy, Tailwind CSS* Aug 2024 – Sept 2024

- Developed a full-stack web application that allows students to upload, process, and share past exam papers
- Built a Flask backend with SQLAlchemy for secure file storage and dynamic PDF generation, integrating it with a React.js frontend for a responsive and intuitive user experience
- Collaborated using Git/GitHub for version control and code reviews, gaining experience using UNIX tools and shell scripting to interact with the operating system

LEADERSHIP EXPERIENCE

Computer Science Student Union @ UofT Sept 2025 – May 2026
Academic Associate Toronto, ON

- Taught peers core data structures and algorithms in Python through presentations and coding examples
- Solved coding problems in real time by optimizing algorithms through problem-solving and Object-Oriented Programming (OOP)
- Organized weekly LeetCode Nights for 50+ students in collaboration with IEEE UofT and UTMIST

Recess Hacks Hackathon April 2025 – Aug 2025
Event Organizer Remote

- Researched, wrote, and sent emails to monetary sponsors, judges, and potential vendors
- Organized a 36-hour student-run hackathon with 150+ participants and \$20,000+ in prizes from 13 companies

ACHIEVEMENTS

Fermat Math Contest (2024) | *Top 1% among 19,000+ contestants internationally* University of Waterloo
Canadian Girls' Math Contest (2024) | *3rd place, Senior Division* CGMC

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Haskell, SQL, Java, C/C++, HTML/CSS
Frameworks & Libraries: React, React Native, Node.js, Flask, Prisma, SQLAlchemy, Expo
ML / AI & Data: PyTorch, XGBoost, Stable Baselines3, OpenCV, pandas, NumPy, Matplotlib, MCP, Power BI
Databases: MySQL, PostgreSQL, MongoDB, SQLite, Supabase
Developer Tools & DevOps: Docker, Git/GitHub, GitHub Actions (CI/CD), Jest, Coveralls, Playwright, SLURM/HPC, Google Colab, Figma, Hugging Face Spaces, UNIX/Shell